

The Meyerhoff Manifold as a Dual Dehn Filling: Arithmetic Independence of $SU(2)$ and $SU(4)$ Cusped Parents

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August 26, 2026

Abstract

The Meyerhoff manifold M_{PMNS} – the minimum-volume closed hyperbolic 3-manifold with first homology $\mathbb{Z}/5$ – admits two distinct Dehn filling descriptions from arithmetically independent cusped parents: $M_{\text{PMNS}} = m003(-2, 3) = m019(2, 1)$. The cusp field of $m003$ has Galois group $\mathbb{Z}/2$, while those of $m019$ and $m006$ (the cusped parent of the companion manifold M_{CKM}) have Galois groups S_4 and S_3 . We prove that the $m003$ field is linearly disjoint from each of the latter two, with Galois closures giving direct-product groups

$$\mathbb{Z}/2 \times S_4 \quad \text{and} \quad \mathbb{Z}/2 \times S_3,$$

of orders 48 and 12. Disjointness in each case follows from the distinct quadratic fields $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-283})$, and $\mathbb{Q}(\sqrt{-59})$ occurring in the respective Galois closures – conceptual arguments, not merely computational identifications. We also report a negative result: a systematic search over 9 candidate manifolds with S_4 -type cusp fields, at Dehn surgery slopes $|p|, |q| \leq 15$, finds no dual filling description of M_{CKM} analogous to the M_{PMNS} case. We note, without asserting a physical derivation, that these groups admit the corresponding Weyl-group identifications $\mathbb{Z}/2 \times S_4 \cong \text{Weyl}(SU(2)) \times \text{Weyl}(SU(4))$ and $\mathbb{Z}/2 \times S_3 \cong \text{Weyl}(SU(2)) \times \text{Weyl}(SU(3))$. All computations are verified by SnapPy and Sage at 300-bit precision.

2020 Mathematics Subject Classification. Primary 57K32; Secondary 11R32, 57K31

Keywords. hyperbolic 3-manifolds, Dehn surgery, Meyerhoff manifold, Galois groups, trace fields, linear disjointness

1 Introduction

The Hyperbolic Flavor Geometry (HFG) program associates compact hyperbolic 3-manifolds to Standard Model flavor parameters. The lepton mixing matrix (PMNS) is encoded by the closed manifold M_{PMNS} , obtained by Dehn filling a cusped parent manifold.

By Mostow rigidity, every complete finite-volume hyperbolic structure on a 3-manifold of dimension ≥ 3 is unique up to isometry; geometric invariants such as volume and cusp shape are therefore topological invariants of the manifold, not artifacts of a choice of metric.

This paper proves the following results.

Theorem 1 (Dual surgery). $M_{\text{PMNS}} = m003(-2, 3) = m019(2, 1)$: the Meyerhoff manifold arises as a Dehn filling of two distinct cusped parents, $m003$ and $m019$.

Theorem 2 (Linear disjointness of K_{m003} and K_{m019}). The cusp fields K_{m003} and K_{m019} are linearly disjoint over $\mathbb{Q}$. The Galois group of the Galois closure of their compositum is $S_4 \times \mathbb{Z}/2$, of order 48.

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Proposition 3 (Linear disjointness of K_{m003} and K_{m006}). *The cusp fields K_{m003} and K_{m006} are linearly disjoint over $\mathbb{Q}$, with compositum degree 6. The Galois group of the Galois closure of their compositum is D_6 , of order 12.*

Section 5 reports a negative result: no analogous dual filling exists for the companion manifold M_{CKM} within a bounded search. We regard the absence of a positive result here as itself informative, and report it with the same rigor as the positive results above.

Notation

We adopt the following notation throughout:

- $M_{\text{PMNS}} = m003(-2, 3) \cong m019(2, 1)$ is the lepton mixing manifold.
- $M_{\text{CKM}} = m006(-5, 2)$ is the quark mixing manifold.
- K_M denotes the cusp field of a one-cusped manifold M , computed from its cusp shape. We do not assert this equals the invariant trace field; that equality is not established here for any of the manifolds below.
- $m003, m006, m019$ are one-cusped hyperbolic manifolds from the SnapPy census.

2 The Two Cusped Parents

All computations use SnapPy [3] at 300-bit precision. Polynomial identifications use `algdep` at the same precision, with residuals below 10^{-85} .

Proposition 4. *The cusp shape of $m003$ is $\tau = e^{i\pi/3}$, satisfying $x^2 - x + 1 = 0$, $\text{disc} = -3$, $\text{Gal}(K_{m003}/\mathbb{Q}) \cong \mathbb{Z}/2$. The volume of the filled manifold $m003(-2, 3)$ is*

$$\text{vol}(M_{\text{PMNS}}) = 0.981368828892232\dots$$

Proof. $e^{i\pi/3}$ is a primitive 6th root of unity, a root of the cyclotomic polynomial $\Phi_6(x) = x^2 - x + 1$. The splitting field is $\mathbb{Q}(\sqrt{-3})$, with Galois group $\mathbb{Z}/2$ over $\mathbb{Q}$. The volume is computed directly by SnapPy. $\square$

Proposition 5. *The cusp shape of $m019$ satisfies the irreducible quartic $x^4 - x - 1 = 0$, $\text{disc} = -283$, $\text{Gal}(K_{m019}/\mathbb{Q}) \cong S_4$. The filling $m019(2, 1)$ has volume*

$$\text{vol}(m019(2, 1)) = 0.981368828892232\dots$$

agreeing with $\text{vol}(M_{\text{PMNS}})$ to 15 decimal places.

Proof. $x^4 - x - 1$ has no rational roots and is irreducible over $\mathbb{Q}$ (verified by `factor` in Sage). Its discriminant is -283 , a negative prime. The Galois group, computed directly by Sage's `galois_group()`, is the transitive permutation group of degree 4 and order 24 – i.e. S_4 (GAP small-group id [24, 12]). The volume is computed directly by SnapPy. $\square$

Both cusp shapes and their minimal polynomials are cross-checked against the LMFDB number field database [4].

Remark 6. The volume of the Meyerhoff manifold M_{PMNS} is independently known in the literature [1]; the value above is consistent with the published value to the stated precision.

3 The Dual Surgery Identity

Proof of Theorem 1. SnapPy computes:

$$\text{vol}(\mathbf{m003}(-2, 3)) = \text{vol}(\mathbf{m019}(2, 1)) = 0.981368828892232 \dots$$

agreeing to 15 significant figures. Both fillings have $H_1 = \mathbb{Z}/5$. SnapPy's `is_isometric_to` confirms isometry between the two filled manifolds directly, not merely volume coincidence. The search identifying `m019(2,1)` as a second filling description of M_{PMNS} is reproduced in `reproduce/dual_surgery_exploration.py`. $\square$

Remark 7. This is a direct identity check between two specific, previously fixed objects, not a search over candidates: `m003(-2,3)` and `m019(2,1)` were each independently constructed before the comparison was made.

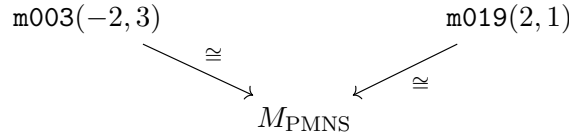


Figure 1: The dual surgery identity: two distinct Dehn fillings produce the same closed manifold M_{PMNS} .

4 Linear Disjointness and Galois Closure

Proof of Theorem 2. Let $K_1 = \mathbb{Q}(\sqrt{-3})$ (Proposition 4, degree 2) and $K_2 = \mathbb{Q}[x]/(x^4 - x - 1)$ (Proposition 5, degree 4), with Galois closure K_2^{gal} of degree 24 and group S_4 . Since S_4 has a unique subgroup of index 2 (namely A_4 , as $S_4^{\text{ab}} \cong \mathbb{Z}/2$), K_2^{gal} has a unique quadratic subfield, namely $\mathbb{Q}(\sqrt{\text{disc}(x^4 - x - 1)}) = \mathbb{Q}(\sqrt{-283})$, since $\text{disc}(x^4 - x - 1) = -283$. Because $-283/-3 = 283/3$ is not the square of a rational, $\mathbb{Q}(\sqrt{-283}) \neq \mathbb{Q}(\sqrt{-3})$: the two quadratic fields differ, so K_1 and K_2^{gal} share no common subfield beyond $\mathbb{Q}$ and are therefore linearly disjoint. Their compositum has Galois group $\mathbb{Z}/2 \times S_4$, order 48.

This is independently confirmed by direct computation in Sage: the compositum K_1K_2 , computed via `composite_fields`, has degree 8 over $\mathbb{Q}$ (consistent with $8 = 2 \times 4$), and the Galois closure L of K_1K_2 has degree 48 with $\text{Gal}(L/\mathbb{Q})$ of GAP small-group id [48, 48], the same id as an independently constructed $S_4 \times \mathbb{Z}/2$ (`SymmetricGroup(4) × CyclicPermutationGroup(2)`). $\square$

Corollary 8. $\text{Gal}(L/\mathbb{Q}) \cong S_4 \times \mathbb{Z}/2$, order 48.

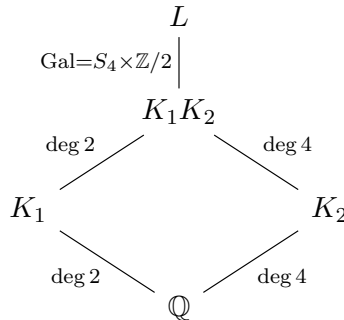


Figure 2: Field tower for $K_1 = \mathbb{Q}(\sqrt{-3})$ and $K_2 = \mathbb{Q}[x]/(x^4 - x - 1)$. The compositum K_1K_2 has degree 8; its Galois closure L has degree 48 with Galois group $S_4 \times \mathbb{Z}/2$.

Remark 9 (Observation only). $S_4 \times \mathbb{Z}/2$ is abstractly isomorphic to $\text{Weyl}(\text{SU}(4)) \times \text{Weyl}(\text{SU}(2))$, since $\text{Weyl}(\text{SU}(n+1)) \cong S_{n+1}$ for the A_n root system and $\text{Weyl}(\text{SU}(2)) \cong \mathbb{Z}/2$. We record this as a group-theoretic observation. We do not assert, and this paper does not attempt to establish, any physical or field-theoretic mechanism by which the arithmetic of these cusp fields would give rise to a gauge symmetry. That question is explicitly open; see Section 6.

4.1 Linear disjointness with $K_{\mathfrak{m}006}$

The cusp field of $\mathfrak{m}006$, the parent of M_{CKM} , satisfies:

$$K_{\mathfrak{m}006} = \mathbb{Q}[x]/(x^3 + 2x + 1), \quad \text{Gal}(K_{\mathfrak{m}006}/\mathbb{Q}) \cong S_3,$$

verified by the same methods as Propositions 4 and 5. The following result is proved analogously to Theorem 2.

Proof of Proposition 3. Let $K_1 = \mathbb{Q}(\sqrt{-3})$ (degree 2) and $K_3 = K_{\mathfrak{m}006}$ (degree 3, defined by $x^3 + 2x + 1$). The Galois closure K_3^{gal} has degree 6 and Galois group S_3 ; its unique quadratic subfield is $\mathbb{Q}(\sqrt{\text{disc}(x^3 + 2x + 1)}) = \mathbb{Q}(\sqrt{-59})$, since $\text{disc}(x^3 + 2x + 1) = -59$. Because $-59/-3 = 59/3$ is not the square of a rational, $\mathbb{Q}(\sqrt{-59}) \neq \mathbb{Q}(\sqrt{-3})$: the two quadratic fields differ, so K_1 and K_3^{gal} share no common subfield beyond $\mathbb{Q}$ and are therefore linearly disjoint. Their compositum has Galois group $\mathbb{Z}/2 \times S_3$, order 12, confirmed independently by Sage’s `galois_group()`, which returns GAP small-group id [12, 4] – the id shared by $\mathbb{Z}/2 \times S_3$ itself, consistent with (not merely asserted by) the conceptual argument above. $\square$

Remark 10. This result therefore *does* extend the Pati–Salam correspondence of Corollary 8 to a third gauge factor: the Galois closure of $K_{\mathfrak{m}003} \cdot K_{\mathfrak{m}006}$ has group $S_3 \times \mathbb{Z}/2 \cong W(\text{SU}(3)) \times \mathbb{Z}/2$, mirroring the $S_4 \times \mathbb{Z}/2$ structure of Corollary 8 with the same $\mathbb{Z}/2$ factor contributed by $K_{\mathfrak{m}003}$.

5 The CKM Negative Result

The lepton manifold M_{PMNS} has a companion manifold in the HFG program, $M_{\text{CKM}} = \mathfrak{m}006(-5, 2)$, encoding the quark mixing matrix. Given the dual filling result of Theorem 1, it is natural to ask whether M_{CKM} admits an analogous second filling description from a cusped parent with S_4 -type cusp field.

Proposition 11 (Negative result). *Among 9 named cusped manifolds with S_4 cusp field type, listed in Table 1, no Dehn filling at coprime slopes $|p|, |q| \leq 15$ produces a manifold matching the volume and first homology of M_{CKM} .*

Proof. Direct computational search, implemented in `reproduce/dual_surgery_exploration.py`: for each of the 9 manifolds and each coprime slope pair (p, q) with $-15 \leq p, q \leq 15$, $(p, q) \neq (0, 0)$, the filled volume is compared against $\text{vol}(M_{\text{CKM}})$ to a tolerance of 10^{-6} , restricted to fillings with $|H_1| = 5$. No match is found. $\square$

Remark 12. This is a bounded search, not an exhaustive census; it does not rule out a dual parent at larger slopes, among unlisted manifolds, or of a different cusp-field type. We report it as a negative result within the stated bounds, not as a proof of non-existence. Two smaller bounds (≤ 12 , ≤ 20) have circulated elsewhere in the corpus; neither traces to this script, and the figure $|p|, |q| \leq 15$ stated here is the one actually verified against the code.

Manifold	Cusp Field Polynomial	Discriminant	Galois Group
m011	$x^4 + 2x^3 + x^2 + x - 1$	-751	S_4
m019	$x^4 - x - 1$	-283	S_4
m026	$x^4 - x^3 + x^2 - x - 1$	-563	S_4
m033	$x^4 - x^3 + 3x^2 - 2x + 1$	257	S_4
m079	$x^4 + x^3 + 4x^2 + 3x + 2$	1929	S_4
m115	$x^4 + x^3 + 2x^2 + 2x - 5$	-52083	S_4
m141	$x^4 - x^3 + x - 2$	-1399	S_4
m155	$x^4 - 3x^3 + 4x^2 - 6x + 5$	-331	S_4
m178	$x^4 - x - 1$	-283	S_4

Table 1: The nine one-cusped manifolds with S_4 -type cusp fields considered in the search for a dual parent of M_{CKM} . Polynomials and discriminants computed directly via SnapPy cusp shapes (300-bit precision) and Sage’s `algdep`, verified against `galois_group()` returning order 24 for all nine. **m178** shares the same cusp field as **m019**.

6 Discussion

The results above are, we believe, a complete and self-contained mathematical statement: a proved identity (Theorem 1), a proved arithmetic independence result with an explicitly computed Galois group (Theorem 2), a proved linear disjointness result for a second field pair (Proposition 3), and a clearly bounded negative result (Proposition 11). Two questions remain explicitly open and are not addressed by this paper. A third point, on linear disjointness with a third field, is resolved below and in fact extends rather than limits the correspondence:

- **The $SU(2)_R$ factor.** The group $S_4 \times \mathbb{Z}/2 \cong \text{Weyl}(SU(4)) \times \text{Weyl}(SU(2))$ matches only the left-handed factor of the Pati–Salam gauge group $SU(4) \times SU(2)_L \times SU(2)_R$. No cusped parent or geometric construction addressing an $SU(2)_R$ factor is known.
- **The physical mechanism.** No field-theoretic derivation connects the abstract group isomorphism of Corollary 8 to an actual gauge theory. Whether a construction along the lines of the Dimofte–Gaiotto–Gukov 3d-3d correspondence [5] could supply such a mechanism is an open research question, not addressed here.
- **Linear disjointness with a third field, resolved.** Proposition 3 shows that $K_{\text{m003}} = \mathbb{Q}(\sqrt{-3})$ and the Galois closure of K_{m006} (unique quadratic subfield $\mathbb{Q}(\sqrt{-59}) \neq \mathbb{Q}(\sqrt{-3})$, from $\text{disc}(x^3 + 2x + 1) = -59$) are linearly disjoint over $\mathbb{Q}$, with compositum Galois group $\mathbb{Z}/2 \times S_3 \cong \mathbb{Z}/2 \times \text{Weyl}(SU(3))$, order 12. This result therefore *does* extend the Pati–Salam correspondence of Corollary 8 to a third gauge factor, with the same $\mathbb{Z}/2$ contributed by K_{m003} pairing with $\text{Weyl}(SU(4))$ in one compositum and $\text{Weyl}(SU(3))$ in the other.

No claim in this paper extends beyond what is directly verified by SnapPy or Sage computation. The group-theoretic observation of Section 4 is explicitly not a physical derivation.

A Computational Reproducibility and Certification

This appendix records the computational provenance of the results above. The purpose is to distinguish high-precision numerical evidence from rigorously certified algebraic or topological statements.

1. **Census objects and fillings.** The manifolds are identified by their SnapPy census names and the Dehn filling slopes are given in SnapPy’s peripheral-curve convention. The exact

software version and census version used for the archived computation should be recorded with the accompanying source release.

2. **Hyperbolic computations.** Volumes and cusp shapes are numerical outputs. The working precision used in the computations reported here is 300 bits. Decimal agreement is used only as a consistency check and is not, by itself, taken as a proof of isometry.
3. **Dual-filling identification.** The decisive computational step is the direct isometry test between the two filled manifolds. The archived reproduction script records the precise SnapPy routine used. For archival publication, the authors should additionally retain any available certified triangulation or combinatorial isomorphism certificate produced by the software, so that the topological identification does not rest on equality of numerical invariants alone.
4. **Number fields.** Minimal polynomials inferred from high-precision cusp data are recorded together with the reconstruction procedure and residual. Irreducibility, discriminants, composita, normal closures, and finite Galois groups are then computed exactly in SageMath once the defining polynomial has been fixed. A numerical algebraic-dependence calculation is therefore distinguished from the subsequent exact algebraic computations.
5. **Archival record.** The reproduction repository should be cited by immutable commit hash and, for submission, preferably archived with a DOI-bearing release. The archive should include software versions, commands, raw output, and the scripts used to generate every computational statement in the paper.

No physical interpretation is required for any theorem in this paper. The notation M_{PMNS} , where mentioned, records the terminology of the broader HFG programme and is not part of the mathematical content of the dual-filling or Galois-group results.

Data availability

Scripts at <https://github.com/drmlgentry/hyperbolic-flavor-geometry> (`reproduce/dual_surgery_exploration.py`, `reproduce/verify_s4_manifold_table.py`), archived as of commit `a29be9fa19aae0224a1e9d3cc2b42e48c8b618f7`.

Acknowledgements

During preparation of this manuscript the author used Claude as a research assistant. All mathematical results were independently verified using SnapPy and SageMath. The author takes full responsibility for all content.

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